COMP262 - Phase 2: Machine Learning Sentiment Analysis

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Project Overview

This report was generated from previously saved model artifacts, figures, and evaluation outputs. It summarizes dataset preparation, sentiment classification model performance, and rating enhancement analysis.

Methodology and Preprocessing

- A balanced real subset was selected from the Amazon Gift Cards review dataset so that the model evaluation was not dominated by the positive class.

- The review text was lowercased to normalize vocabulary and reduce duplicate tokens that differ only by casing.

- Contractions such as n't were expanded so negation could be captured more reliably during preprocessing.

- Negation handling was applied so short windows after words like not and no were marked with a NOT_ prefix, helping the models distinguish phrases such as not good from good.

- Non-alphabetic characters were removed while preserving exclamation marks because they can signal stronger sentiment intensity.

- Stopwords were removed to reduce noise, but key negation words were retained so sentiment polarity was not lost.

- Lemmatization was used to reduce inflected forms to a common base word and improve generalization.

Text Representation Choice

TF-IDF was selected because it provides a strong and interpretable baseline for review classification, works well with Naive Bayes and MLP, and highlights informative unigrams and bigrams without requiring a large neural language model for feature extraction.

Dataset Information

Metric	Value
Full dataset rows	147194
Balanced subset rows	8988
Training rows	6291
Test rows	2697
Subset class distribution	{'Negative': 2996, 'Positive': 2996, 'Neutral': 2996}
Subset rating distribution	{'1': 2450, '2': 546, '3': 2996, '4': 186, '5': 2810}
Training class distribution	{'Negative': 2097, 'Neutral': 2097, 'Positive': 2097}
Test class distribution	{'Negative': 899, 'Neutral': 899, 'Positive': 899}

Training rating distribution	{'1': 1715, '2': 382, '3': 2097, '4': 130, '5': 1967}
Test rating distribution	{'1': 735, '2': 164, '3': 899, '4': 56, '5': 843}

Model Evaluation Summary

Model	Accuracy	Precision	Recall	F1
VADER	0.4405	0.4824	0.4405	0.3854
TextBlob	0.4761	0.4804	0.4761	0.4522
Naive Bayes	0.7063	0.7019	0.7063	0.7026
MLP	0.7104	0.7079	0.7104	0.7084

Training Process Results

- Naive Bayes best parameters: {'alpha': 1.0}

- Naive Bayes best cross-validation weighted F1: 0.6955

- MLP best parameters: {'alpha': 0.01, 'hidden_layer_sizes': [64, 32]}

- MLP best cross-validation weighted F1: 0.6975

- The training process used GridSearchCV with stratified folds to tune each model before final evaluation on the held-out 30% test set.

Lexicon Versus Machine Learning Experiment Design

To ensure an apples-to-apples comparison, the lexicon-based methods and the machine learning models were all evaluated on the exact same held-out test split. The gold labels came from the rating-derived sentiment field, and VADER, TextBlob, Naive Bayes, and MLP were then scored with the same accuracy, precision, recall, and weighted F1 metrics.

Rating Enhancement Using Review Text

One option suggested by the recommender-systems literature is to refine a user rating by blending the original star value with a sentiment score estimated from the review text. In this project, the VADER compound score is normalized to the same range as the original star rating and then combined with the original score using a weighted average.

Diagram

```
Original review text -> VADER sentiment score -> normalize to [1, 5]
                                  |
                                  v
Original star rating --------> weighted blend --------> enhanced rating
```

Pseudocode

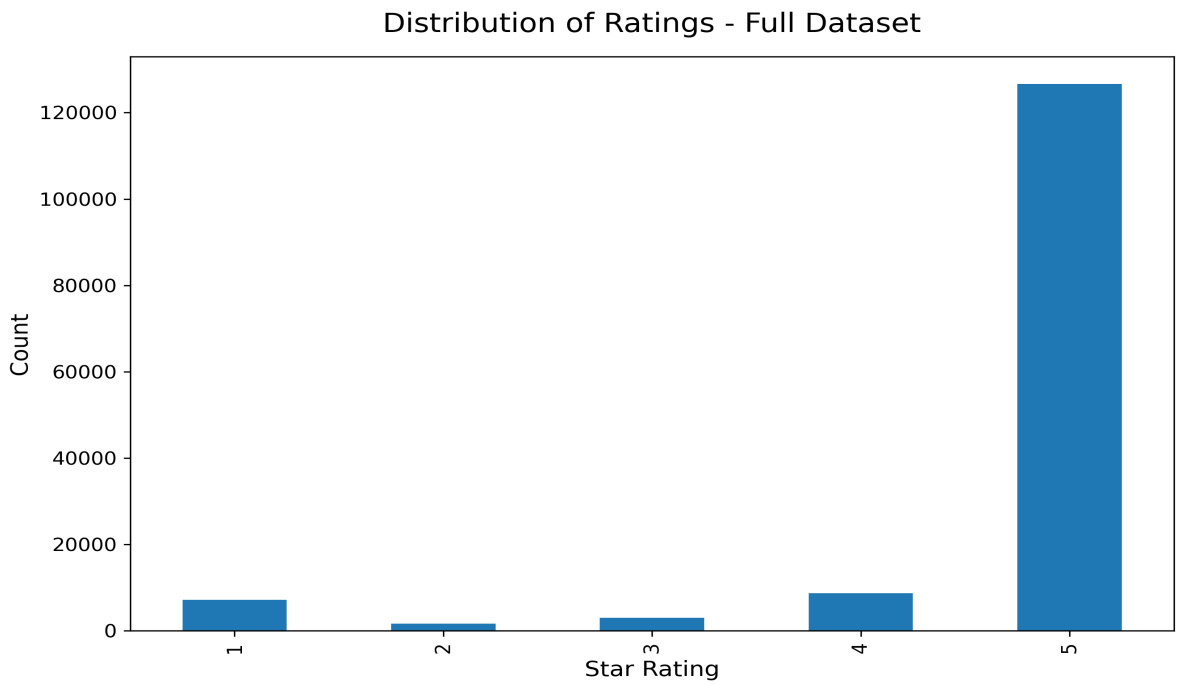
```
for each review:
    compound = vader(review_text).compound
```

```
normalized_sentiment = ((compound + 1) / 2) * 4 + 1
normalized_star = original_star
enhanced_rating = alpha * normalized_star + (1 - alpha) * normalized_sentiment
save enhanced_rating
```

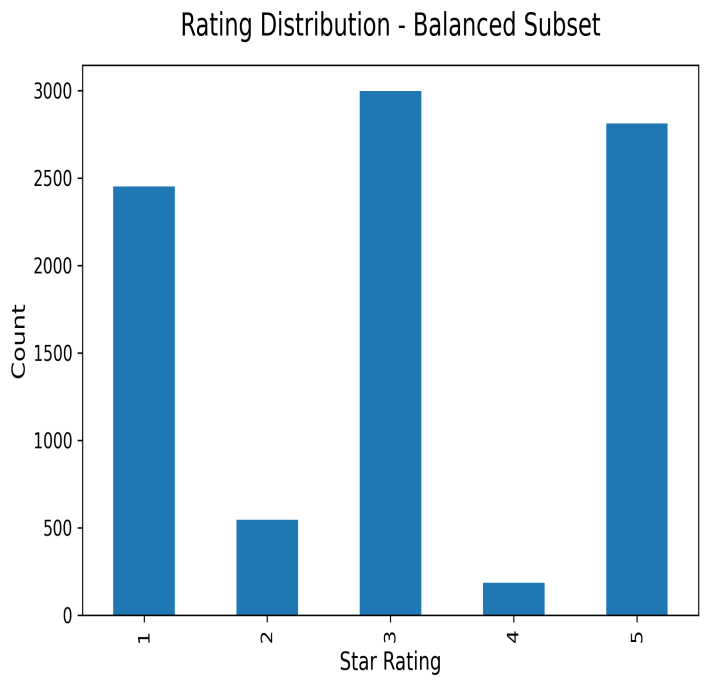
The saved enhanced-rating outputs show how textual sentiment can reduce or reinforce the original star value. Large disagreements indicate reviews where the text tone and rating are not well aligned.

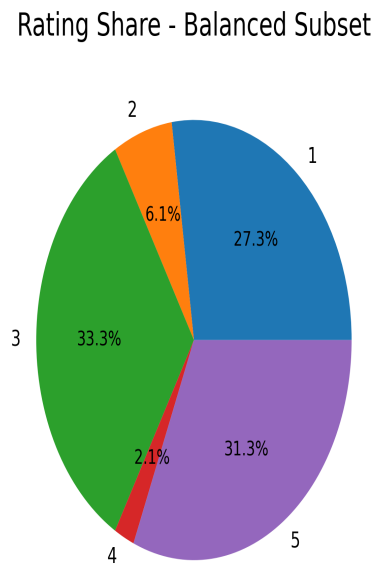
Figures

Distribution of Ratings - Full Dataset

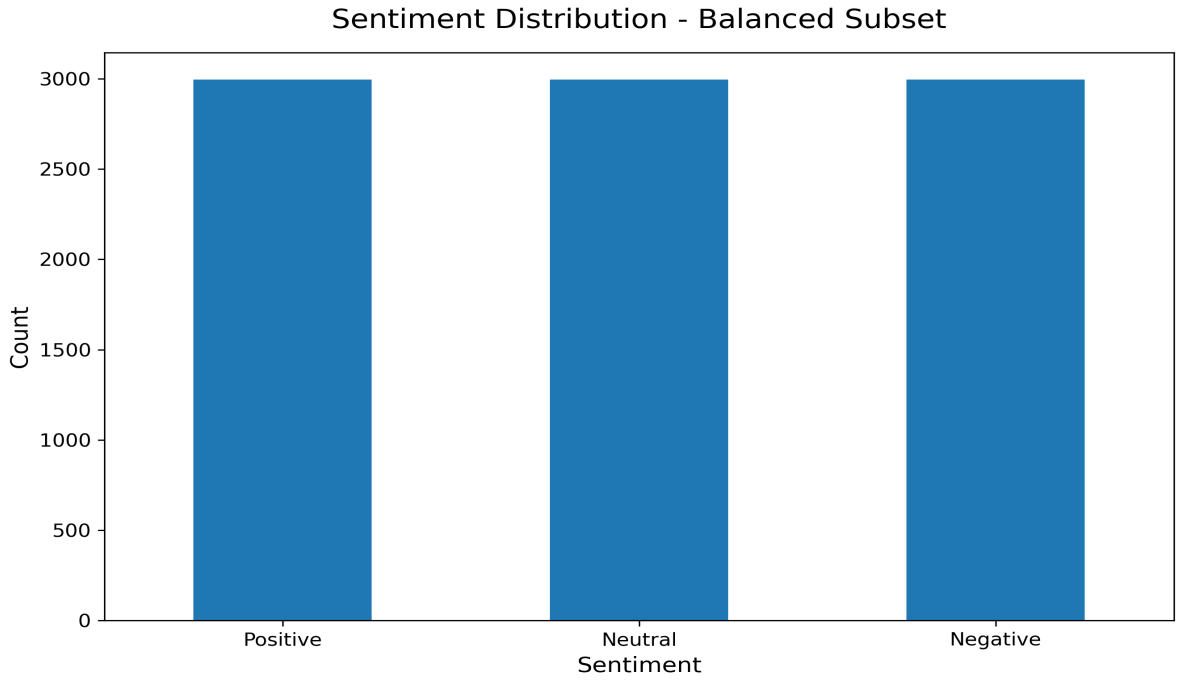


Rating Distribution - Balanced Subset

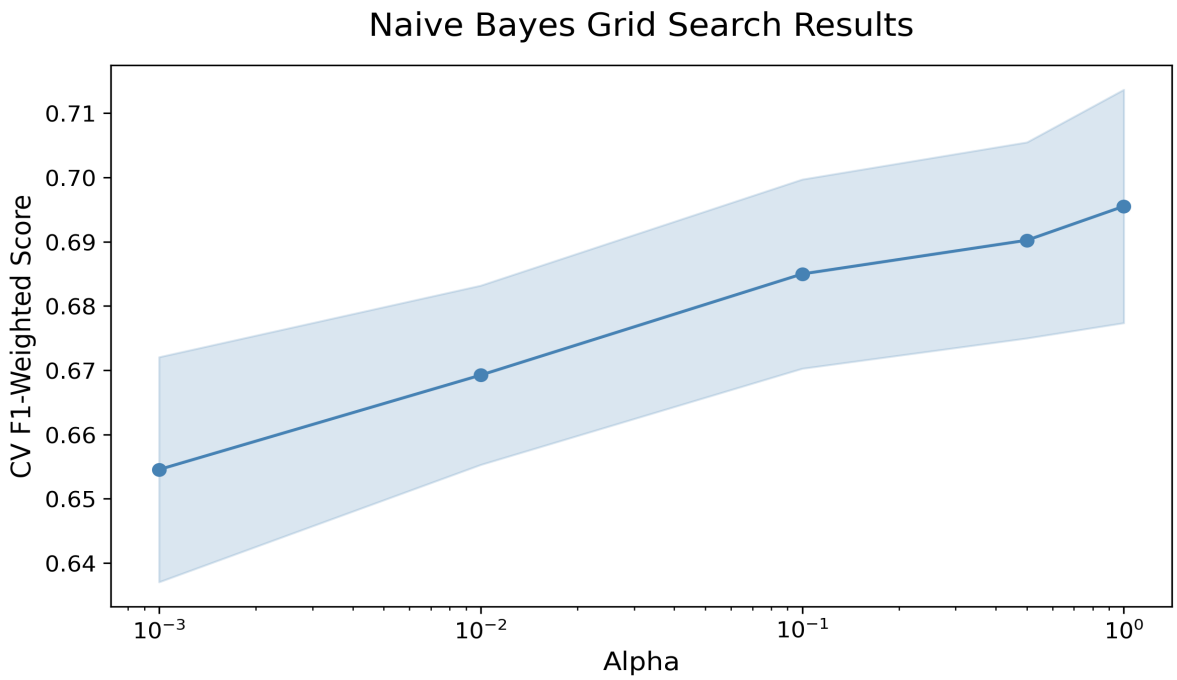




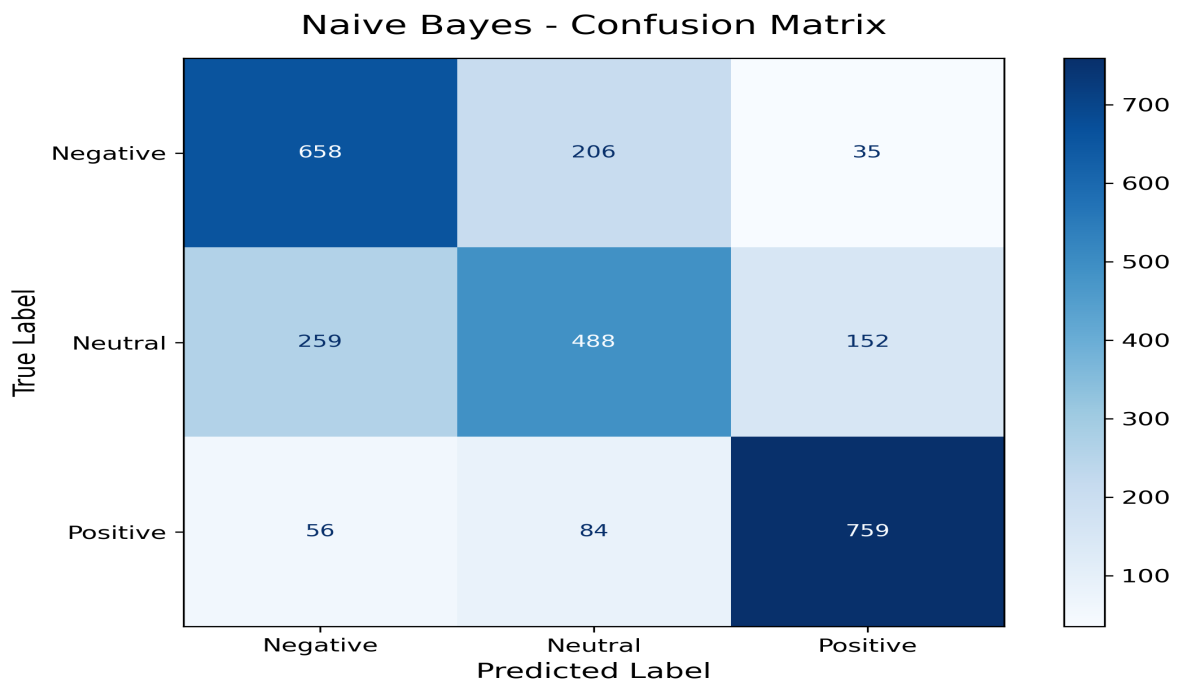
Sentiment Distribution - Balanced Subset



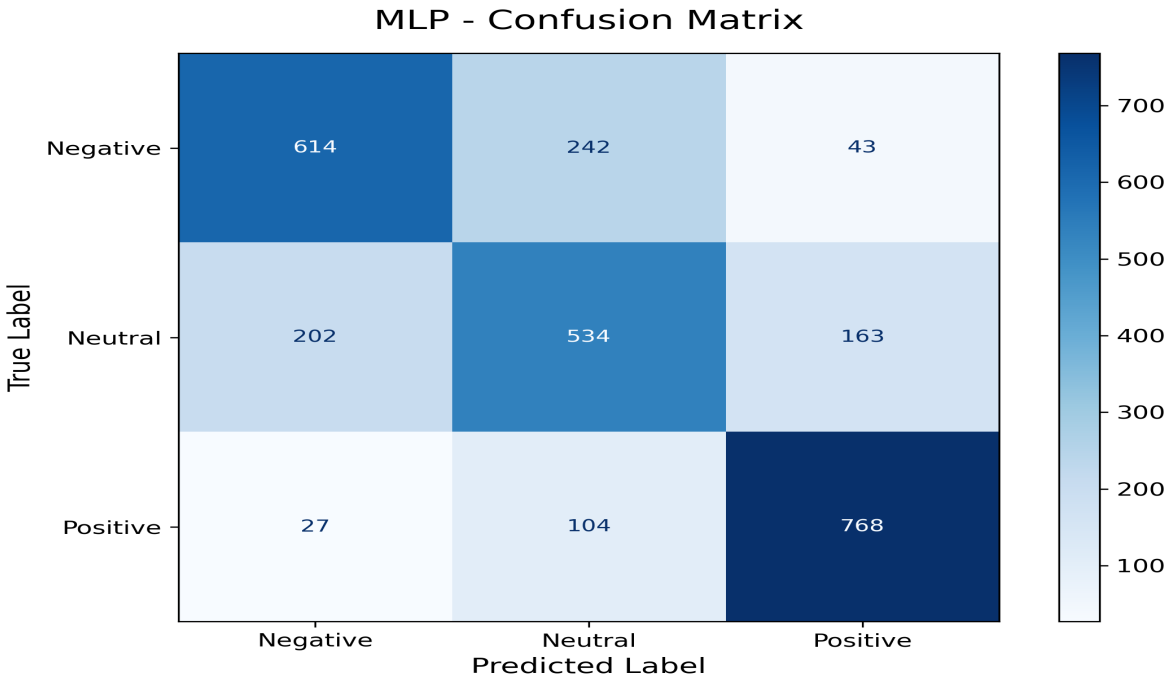
Naive Bayes Grid Search Results



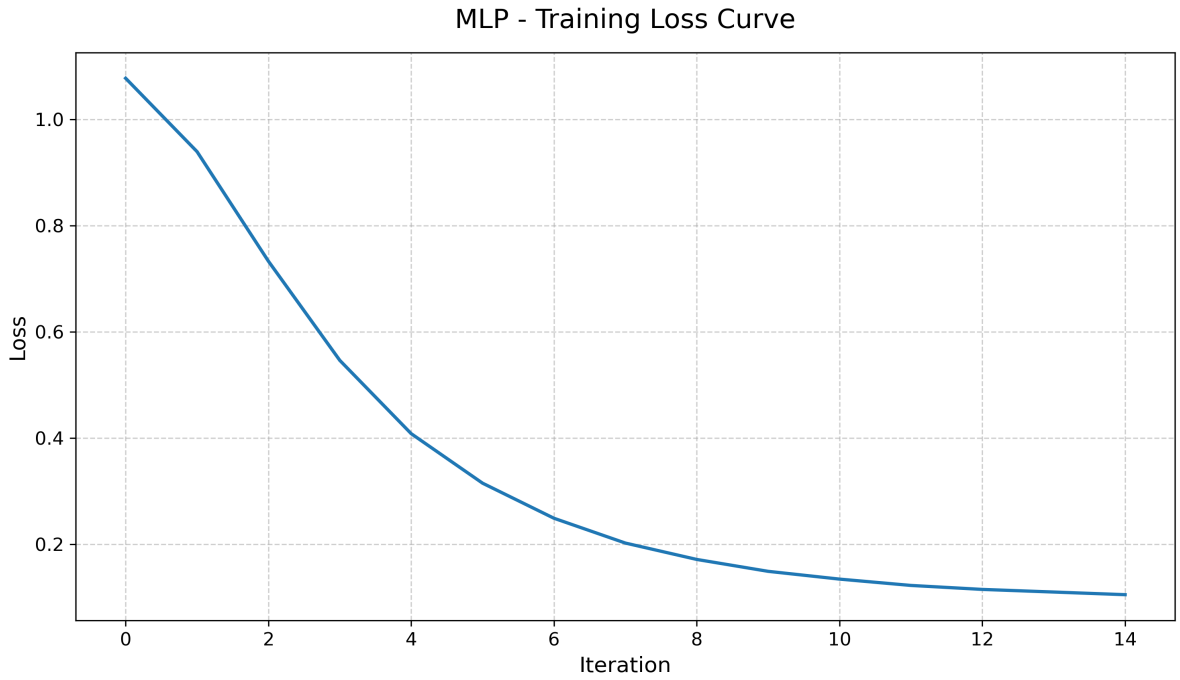
Naive Bayes Confusion Matrix



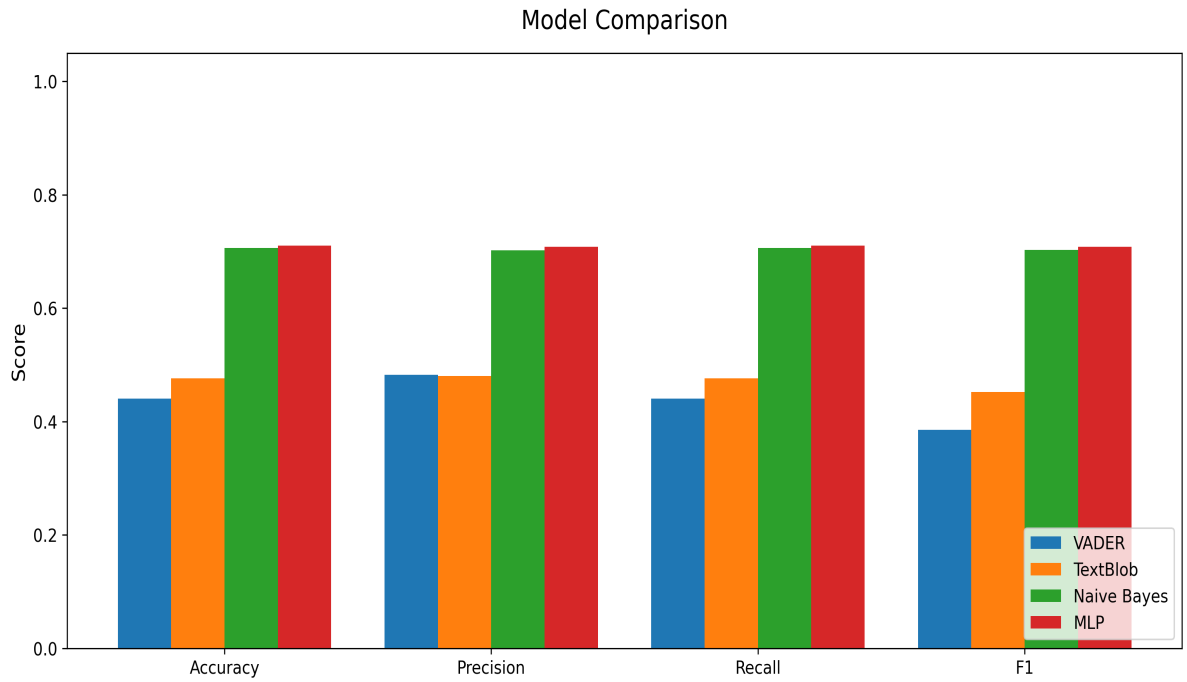
MLP Confusion Matrix



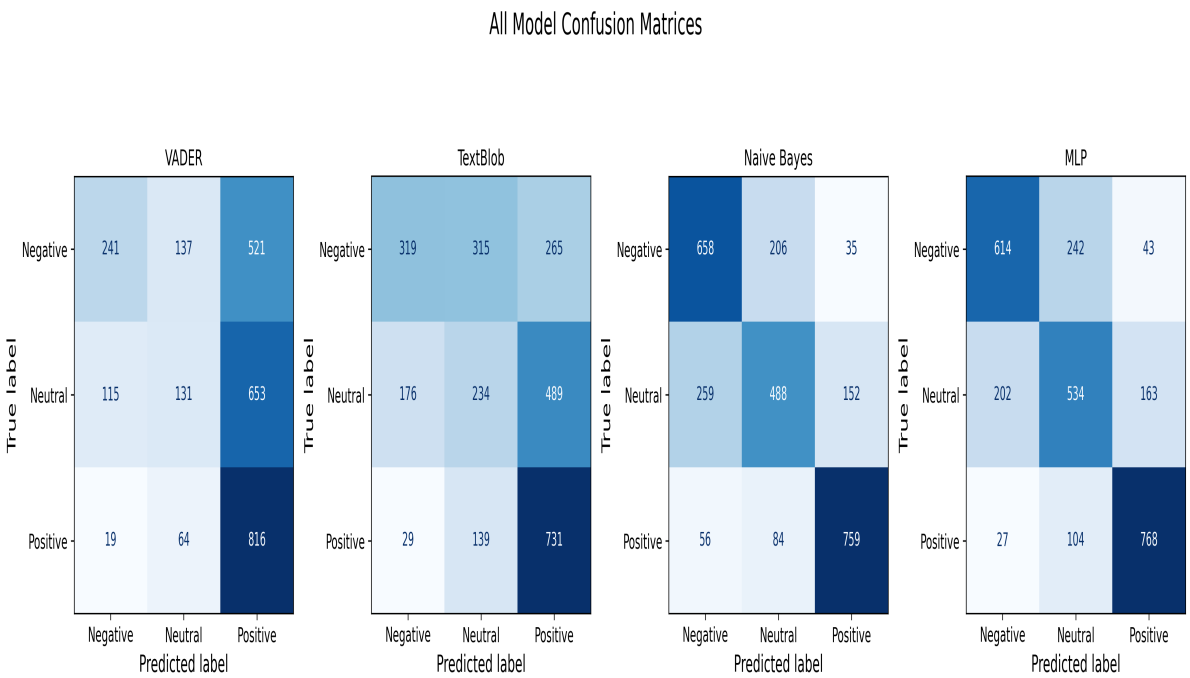
MLP Training Loss Curve



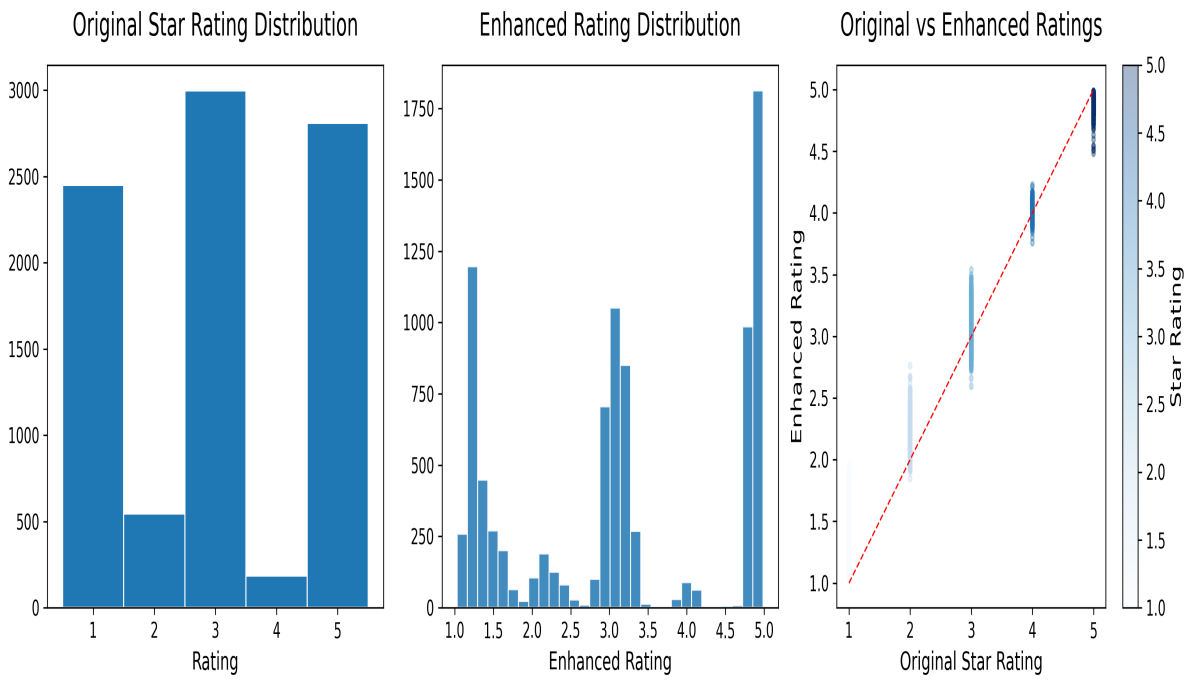
Model Comparison Bar Chart



All Model Confusion Matrices



Enhanced Rating Analysis


Original Star Rating Distribution
Rating
Enhanced Rating Distribution
Enhanced Rating
Original vs Enhanced Ratings
Original Star Rating
Enhanced Rating
Star Rating

Classification Reports

VADER

```
precision    recall  f1-score   support

    Negative       0.64      0.27      0.38       899
     Neutral       0.39      0.15      0.21       899
    Positive       0.41      0.91      0.56       899

    accuracy                           0.44      2697
   macro avg       0.48      0.44      0.39      2697
weighted avg       0.48      0.44      0.39      2697
```

TextBlob

```
precision    recall  f1-score   support

    Negative       0.61      0.35      0.45       899
     Neutral       0.34      0.26      0.29       899
    Positive       0.49      0.81      0.61       899

    accuracy                           0.48      2697
   macro avg       0.48      0.48      0.45      2697
weighted avg       0.48      0.48      0.45      2697
```

Naive Bayes

```
precision    recall  f1-score   support

    Negative       0.68      0.73      0.70       899
     Neutral       0.63      0.54      0.58       899
    Positive       0.80      0.84      0.82       899

    accuracy                           0.71      2697
   macro avg       0.70      0.71      0.70      2697
weighted avg       0.70      0.71      0.70      2697
```

MLP

```
precision    recall  f1-score   support

    Negative       0.73      0.68      0.70       899
     Neutral       0.61      0.59      0.60       899
    Positive       0.79      0.85      0.82       899

    accuracy                           0.71      2697
   macro avg       0.71      0.71      0.71      2697
weighted avg       0.71      0.71      0.71      2697
```

LLM Tasks

Requirement 16 - Summaries

The summarization model used was Qwen/Qwen2.5-0.5B-Instruct. Ten reviews longer than 100 words were selected, and the first two are shown below.

Review 1 Original Text

I ordered this card, along with another. They shipped the same morning I purchased them. The next morning the tracking on USPS said it was delivered but it was not. I emailed Amazon with the problem and they sent out new cards right away. The new cards arrived the next morning at my door, via UPS. 2 days later, the original order showed up. Inside the box of the original order was a a black box with an orange ribbon gift bow and with Amazon`s logo allover. When I opened the box, there was a white gift card with a snow globe and Amazon`s logo in the middle. It was beautiful and Amazon`s customer service, via e-mail was amazing and lightening fast. Happy Holidays.

Review 1 50-Word Summary

The customer received two identical cards from different shipping companies, which were then incorrectly tracked to be delivered. After emailing Amazon for assistance, the company promptly sent out new cards that arrived the next day. Upon opening the box, they found a beautifully designed gift card with a snow globe and Amazon's logo. The experience was described as "amazing" and "lightning fast." The customer is happy with

Review 2 Original Text

I wanted to purchase a gift card as a wedding present. After reading about the gift card gaffes, I thought I would try the "print at home" version. I have successfully done this a year ago for a baby shower gift. When Amazon sent the email to "print at home," it said I needed to download Adobe Acrobat. I have had trouble with Adobe since its inception. However, I tried. My computer would not allow the download because of issues with McAfee Security. I called (on the phone) Amazon, told them the problems I was having, and they have credited my account and hopefully resolved the issue. They could not have been nicer. This is my first problem with Amazon. I shall continue to be a loyal Amazon shopper, but will stay away from gift card purchases on-line.

Review 2 50-Word Summary

Summary: The reviewer purchased a gift card as a wedding present, initially expecting a print-at-home option. Upon trying the "print at home" version, they encountered difficulties due to software compatibility issues. Despite attempts to resolve these problems through Amazon's support team, the experience did not meet their expectations. The reviewer now considers Amazon's online gift card purchases less appealing than traditional in-store options.

Requirement 17 - Service Representative Response

Selected customer review:

Amazon, I am shopping for Amazon.com gift cards for Christmas gifts and am really so disappointed that out of five choices there isn't one that says "Merry Christmas" or mentions Christmas at all! I am sure I am not alone in wanting a card that reflects the actual "holiday" we are celebrating. On principle, I cannot send a Amazon gift card this Christmas. What's up with all the Political Correctness? Bad marketing decision. Lynn

Generated response:

Thank you for your feedback. We understand how disappointing it can be to receive a gift card without any mention of the holiday season. Our goal is always to provide thoughtful and accurate information about our products. To rectify this issue, we will add a note on the gift card stating "This gift card represents the festive spirit of Christmas." This should help clarify what the gift card is meant to represent. If you have any other questions or need further assistance, please don't hesitate to reach out. We're here to help make your shopping experience as enjoyable as possible.

Key Findings

- Best model by weighted F1: MLP (0.7084)

- Machine learning models outperform lexicon-based methods on the same test set.

- The balanced real subset provides a more reliable evaluation than earlier oversampled experiments.

- Neutral sentiment remains the hardest class, while Positive is generally the easiest to classify.